

- **Student**(SID, Name, Major, Year)
- **Course**(CID, Title, Department, Credits)
- **Instructor**(IID, Name, Department)
- **Enrollment**(SID, CID, Grade)
- **Teaches**(IID, CID, Semester)

Even questions

HW #1 pt. B

$$2) \pi_{\text{Title}} \left(\sigma_{\text{Department} = 'cs'} (\text{Course}) \right)$$

$$4) \pi_{\text{Name}} \left(\text{Instructor} \bowtie_{\text{IID}} \left(\text{Teaches} \bowtie_{\text{CID}} \left(\sigma_{\text{Department} = 'cs'} (\text{Course}) \right) \right) \right)$$

$$6) \pi_{\text{Name}} \left(\text{Student} \bowtie_{\text{SID}} \left(\left(\pi_{\text{SID, CID}} (\text{Enrollment}) \right) \div \left(\pi_{\text{CID}} \left(\sigma_{\text{Department} = 'cs'} (\text{Course}) \right) \right) \right) \right)$$

$$8) \pi_{\text{Title}} \left(\text{Course} \bowtie_{\text{CID}} \left(\pi_{\text{CID}} (\text{Course}) - \pi_{\text{CID}} (\text{Enrollment}) \right) \right)$$

$$10) \pi_{\text{Name}} \left(\text{Instructor} \bowtie_{\text{IID}} \left(\sigma_{\text{count} > 1} \left(\gamma_{\text{IID, count}(\text{CID}) \rightarrow \text{count}} (\text{Teaches}) \right) \right) \right)$$